

Continuous and Closed-Form Methods for Leaky Integrate-and-Fire Archetype Analysis

A Methodology Compendium

What differential-equation lenses see in spiking-network archetypes, and what only discrete verification reaches

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Abstract. The CogSpike programme verifies spiking neural networks **discretely** — the prior paper [1] encodes them as Markov chains and model-checks them in PRISM, and the De Maria et al. “FCS” line [2] encodes leaky integrate-and-fire (LI&F) neurons in Lustre and verifies temporal-logic properties with Kind2. Discrete verification is exact but exponential, and it certifies a property at a parameter point without describing the **shape** of the region where it holds. Over seven research threads we asked whether a **continuous** lens — differential equations, eigenvalues, transfer functions, bifurcation theory — could see structure the discrete lens cannot. This compendium consolidates all seven. The answer is a clean two-sided one. Continuous methods do recover shape and scale: a single eigenvalue predicts the winner-take-all boundary at 99.96% in mean-field; the contralateral pitchfork and the activator–inhibitor Hopf are exact closed-form curves; the Siegert formula gives a 99.6%-recall envelope of the FCS winner-take-all region. But every continuous method systematically **misses the integer-tick spike-timing physics** — the exact staircase of synchronous-lock cells, the exact oscillation period, the binary 1100 waveform, deterministic lock-in. One nonlinearity, the **spike-reset rule**, is responsible for the entire boundary between what these methods reach and what they do not, and whether it appears in the property being verified is a decidable, a-priori test. The six methods form a **ladder** of increasing fidelity and decreasing tractability — linearization, transfer functions, Wilson–Cowan mean-field, the Siegert closed form, quasi-renewal mesoscopics, and finally the discrete FCS oracle itself. This document is organised by that ladder. It reports the successes and the dead ends with equal weight, because the dead ends turned out to be the sharpest result: they measure exactly how much of a verified property is irreducibly discrete.

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1. Introduction and the central finding

1.1. The starting point: the discrete programme

The CogSpike project studies **spiking neural networks** (SNNs): networks of neurons that communicate by discrete, all-or-nothing spikes rather than continuous signals. The neuron model throughout is the **leaky integrate-and-fire** (LI&F) neuron — a unit that sums weighted input into a membrane potential, leaks a fraction of it away each step, and emits a spike (then resets) when the potential crosses a threshold.

The programme’s prior work is **discrete** end to end. The published CogSpike paper [1] encodes a probabilistic SNN as a Discrete-Time Markov Chain and verifies behavioural properties with the PRISM model checker, introducing a weight-discretization scheme to fight the **state-space explosion problem** (the chain grows exponentially with neuron count). The body of work it builds on — De Maria et al., which we abbreviate **FCS** throughout after the journal of its archetypes paper [2] — encodes LI&F neurons in the synchronous language Lustre and verifies temporal-logic properties with the Kind2 model checker. Both are **sound and exact**: they give a machine-checked yes/no answer. Both share two limitations. They are **exponential** in network size, and they verify a property **at a point** in parameter space — one integer weight assignment at a time — without telling you the shape of the region of weights where the property holds, or why.

Plain-language reading. A model checker is like testing a circuit by trying every input combination. It is exhaustive and trustworthy, but it scales badly, and when it finishes you know **that** the circuit works, not **why**, and not what margin you have before it stops working.

1.2. The question

A spiking network is a **dynamical system**: a state that evolves in steps. The mature mathematics of dynamical systems — differential equations, eigenvalues, transfer functions, bifurcation theory — was built for **continuous** systems, and it answers exactly the questions a model checker does not: where is the boundary of a behaviour in parameter space, how fast does the system settle, what happens just past the boundary. The seven-thread research programme consolidated here asked one question:

Can a continuous, differential-equation lens reveal structure in LI&F spiking-network archetypes that the discrete model-checking lens cannot — can we see these systems “as if continuous” and read off shapes invisible to exhaustive verification?

The obstruction is one specific feature of a spiking neuron: the **reset**. The membrane potential integrates smoothly, then — at a spike — jumps discontinuously back down. Calculus, eigenvalues, and every tool above presuppose a **smooth** system. The reset is not smooth. The seven threads are, in effect, seven different ways of getting around the reset, and seven measurements of what that costs.

1.3. The answer

The answer is two-sided and consistent across all seven threads.

Continuous methods recover the shape and scale of behaviour — boundaries, envelopes, bifurcation curves, oscillation existence, reachability — but system-

atically miss the integer-tick spike-timing physics: the exact staircase of synchronous-lock cells, the exact integer period, the binary spike waveform, deterministic lock-in. The two are complementary, not competing. The shapes the project hoped to see are real and useful; the discrete lens nonetheless retains a residual “positive content” that no continuous theory reaches.

Both halves are quantified later, but the headline numbers are worth stating now. On the positive side: a single eigenvalue predicts the winner-take-all boundary at 99.96% accuracy in the mean-field description (Section 3.1); the winner-take-all and oscillation onset curves are derivable in **closed form** and match simulation to machine precision (Section 2.4); the Siegert formula gives an **envelope** of the FCS winner-take-all region with 99.6% recall, usable as a verification pre-filter (Section 2.5). On the negative side: no rate-equation method exceeds a **Jaccard agreement of roughly 0.70** with the discrete FCS winner-take-all oracle — the residual ≈ 0.30 is the integer-tick lock that only discrete verification resolves (Section 3.1); and the predicted oscillation period is wrong by a **structural factor of four** that no recalibration can fix (Section 3.2).

The single mechanism behind every line of that paragraph is the spike-reset rule. This is the **diagnostic principle** of the whole programme, and it recurs in every thread:

Plain-language reading. The spike-reset is the one nonlinearity that every continuous method erases. Whether the reset **matters** for the property you are verifying decides whether continuous methods apply. Properties about the **existence and stability of attractors** — can a winner emerge, is the network stable, does it oscillate at all — do not depend on the exact reset schedule, and continuous methods handle them. Properties about the **exact integer schedule of spikes** — which precise period, which binary pattern, deterministic lock-in — are the reset schedule, and continuous methods cannot reach them.

1.4. How to read this document

The six methods used across the seven threads form a **ladder**, ordered from the cheapest and coarsest to the most faithful and most expensive. The ladder is the spine of this document:

- Section 2 climbs the ladder rung by rung. Each rung gets a plain-language introduction — what the method is, why it exists, an analogy to something a formal-methods reader already knows — **before** any equation. No prior background in differential equations or control theory is assumed. Section 2.1 is a one-page primer and a discrete-to-continuous translation table that the rest of the document leans on.
- Section 3 crosses the ladder the other way. It takes the three phenomena the threads studied — winner-take-all, oscillation, reachability — and walks each one **down** the ladder, so the reader sees every method applied to the same target. This is where the thread-by-thread results live.
- Section 4 is the index: a capability matrix of method $\times$ phenomenon, with the verdict and the one number that supports it in each cell, and an explicit list of what each method **cannot** do.
- Section 5 collects the dead ends — the hypotheses that failed — and argues that each failure, because it falls exactly where the diagnostic principle predicts, is itself a result.
- Section 6 connects the compendium back to the discrete CogSpike paper and forward to a publication.

Three appendices follow: a thread-by-thread chronology (Section 7), per-thread phase scorecards (Section 8), and a glossary of every technical term (Section 9). A reader who meets an unfamiliar word should expect it to be defined in the glossary and introduced in full at its first ladder rung.

2. The methodology ladder

The seven threads use six analysis methods. They are not six competitors for the same job: they form a **ladder**. Each rung makes one more concession to the discrete spike physics, gains fidelity, and loses tractability. Rung 1 is a napkin calculation; rung 6 is the exponential model checker the whole programme is trying to avoid. The art is knowing which rung a given question needs.

This section climbs the ladder. Every rung opens with a plain-language box — what the method is and why it exists — and only then states the equations, each followed by a sentence of intuition. The phenomena and the numbers come in Section 3 this section is the vocabulary.

2.1. A reader’s primer

A **dynamical system** is a rule that advances a **state** one step at a time. For an SNN the state is the vector of membrane potentials, and the rule is “integrate weighted input, leak, and spike-and-reset on threshold crossing”. Everything in this document is an attempt to analyse that rule.

The rule has two parts. The **subthreshold** part — integrate and leak — is **linear**: doubling the input doubles the response. Linear systems are completely understood. The **spiking** part — the threshold test and the reset — is **nonlinear** and, worse, **discontinuous**: at a spike the state jumps. The six methods differ only in how they handle that discontinuity, from “ignore it” (rung 1) to “simulate it exactly” (rung 6).

Five ideas are used throughout. Each is developed properly at its rung; this is the one-paragraph version.

- **Linearization.** Near an operating point, replace the rule by its best linear approximation. The error is small for small deviations.
- **Eigenvalue.** For a linear rule $x \rightarrow Mx$, an eigenvalue λ is a number such that some direction is simply rescaled by λ each step. $|\lambda| < 1$ means that direction shrinks (stable); $|\lambda| > 1$ means it grows. The **spectral radius** $\rho(M) = \max|\lambda|$ is the single number that decides whether the whole system settles.
- **Fixed point.** A state the rule maps to itself — an equilibrium. A network “at rest” sits at a fixed point; **which** fixed point, and whether it is stable, is the whole game.
- **Frequency domain.** Instead of asking how the system relaxes on its own, ask how it responds when **driven** by an input that oscillates at frequency ω . The answer is a single complex number $H(\omega)$ per frequency.
- **Bifurcation.** As a parameter (a synaptic weight) is varied, a fixed point can change its stability or split in two. The parameter value where this happens is a **bifurcation**, and it is the **boundary** of a behaviour — the shape the discrete lens cannot draw.

Because the threads move between the discrete FCS world and the continuous world, every continuous symbol has a discrete counterpart. Table 1 is the translation table; it is used without further comment from here on.

Object	Discrete world (FCS)	Continuous world	One-line intuition
Time	integer tick t	continuous time, or a step of a smooth map	The FCS neuron advances in indivisible ticks; the continuous lens smears them into a flow.
Activity	Boolean spike $y \in \{0, 1\}$	firing rate $\nu \in [0, 1]$	One neuron either spikes or not; a population of them has a continuous fraction firing.
Memory / leak	windowed integrator, r -vector $[10, 5, 3, 2, 1]$	time constant τ	How long past input lingers. The FCS 5-tap window is the discrete form of an exponential leak.
Threshold	integer $\tau_{\text{th}} = 105$, hard	a smooth gain function f or Φ	A single hard threshold becomes a smooth input–output curve once averaged over many neurons.
Weight	integer w (scaled $\times 10$)	real weight w or matrix J	The wiring. Unchanged in meaning; only the arithmetic differs.
The reset	mem taps zeroed after a spike — exact	averaged away, or linearized away	The defining nonlinearity. Rungs 1–5 each lose it in a different way; rung 6 keeps it.
Eigenvalue λ	—	mode growth/decay factor	No discrete analogue: this is new information the continuous lens adds.
Fixed point	an absorbing set / a periodic lock	a root of $x = F(x)$	Where the system comes to rest, or cycles.
Bifurcation	the edge of a verified region on the weight grid	a parameter value where stability changes	The continuous lens draws this edge as a curve; the discrete lens only samples it cell by cell.

Table 1: **The discrete–continuous bridge.** Every continuous symbol used in this document, with its FCS counterpart and a plain reading. The last three rows have no discrete analogue — they are precisely the new structure the continuous lens contributes.

One FCS-specific detail recurs and is worth stating once. An FCS neuron keeps a length-5 memory buffer of recent weighted inputs, combines them with the fixed weights $r = [10, 5, 3, 2, 1]$ into $V(t) = \sum_e r_e \cdot \text{mem}[e](t)$, fires when $V(t) \geq 105$, exports the spike one tick later, and — the crucial part — **zeroes the shifted memory taps on the tick after a spike**. That zeroing is the reset. A neuron driven by constant input of weight ≥ 11 fires on the first tick and is called a **delayer**. The two archetypes studied throughout are **contralateral inhibition** (two neurons,

each inhibiting the other — produces winner-take-all) and the **negative loop** (an activator exciting an inhibitor that feeds back — produces oscillation).

2.2. Rung 1 — Linearization, eigenvalues, spectral radius

What this method is. A leaky neuron’s subthreshold behaviour is a **linear recurrence**: next potential = leak factor $\times$ current potential + weighted input. The whole network, ignoring spikes for a moment, is one linear map iterated: $x \rightarrow Mx + \text{input}$. Linear maps are solved completely by their **eigenvalues**. An eigenvalue λ is a number attached to a direction that the map merely rescales; after t steps that direction has been scaled by λ^t . So $|\lambda| < 1$ directions die out, $|\lambda| > 1$ directions take over, and the largest $|\lambda|$ — the **spectral radius** ρ — decides the network’s fate. This is rung 1: linearize, take eigenvalues, read off stability.

The LI&F membrane recurrence, with leak r , weight matrix W , spike outputs y , and external drive u , is

$$p_i(t+1) = r \cdot p_i(t) + \sum_j W_{ij}y_j(t) + B_i u_i(t). \quad (1)$$

This says each neuron keeps a fraction r of its potential and adds the weighted spikes it received. Strip the spike nonlinearity and Equation 1 is a textbook linear recurrence. To bring the spikes back in **approximately**, replace the Boolean spike y_j by a smooth firing-rate function $f(p_j)$ and linearize around an operating point p^* . The deviation $\delta p = p - p^*$ then obeys

$$\delta p(t+1) = A \cdot \delta p(t), \quad A = rI + W \cdot \text{diag}(f'(p^*)), \quad (2)$$

where A is the **Jacobian** — the best linear approximation of the dynamics at p^* . Its eigenvalues are the whole story: the system is stable exactly when $\rho(A) = \max_i |\lambda_i(A)| < 1$.

Plain-language reading. A formal-methods reader already knows this theorem in another guise. $\rho(A) < 1$ is the convergence test for a fixed-point iteration — the same condition that decides whether $x^{(k+1)} = Mx^{(k)} + c$ converges. And the **eigenvalue gap** (how far the top eigenvalue stands above the rest) is the spiking-network analogue of the **spectral gap** of a Markov chain, which controls how fast a random walk mixes. Eigendecomposition of the weight matrix is, quite literally, PageRank applied to the wiring diagram: the dominant eigenvector names the neuron that wins.

What it costs. An eigendecomposition is $O(n^3)$ — polynomial, cheap, done in milliseconds for the small archetypes here. That is the entire appeal: rung 1 is the napkin calculation that a model checker’s exponential cost is being traded against.

Where it appears. The foundational threads (Section 7, threads 1–2) established rung 1. On a contrived 4-neuron winner-take-all network the dominant eigenvector of W correctly names the winner and the eigenvalue gap of 20.9 quantifies its margin; the linearized $\rho(A) = 0.629$ correctly predicts settling in ≈ 6 steps. The honest test came when this was carried to the real FCS archetypes (Section 3.1, Section 3.3), where it both succeeded and failed — in a way that turned out to be the project’s first major finding.

2.3. Rung 2 — Transfer functions and frequency response

What this method is. Eigenvalues describe how a system relaxes when **left alone**. A **transfer function** describes how it responds when **driven**. Feed a linear system an input

that wiggles at frequency ω ; it responds at the same frequency, but rescaled and delayed. The complex number $H(\omega)$ records that rescaling (its size) and that delay (its angle) at every frequency. The one idea to keep: a leaky neuron is a **low-pass filter** — it follows slow inputs faithfully and ignores fast ones, because the leak averages rapid fluctuations away.

Taking the Z -transform of the linear recurrence Equation 1 turns a single neuron into the transfer function $H(z) = 1/(z - r)$ — a first-order low-pass filter with its “pole” at $z = r$. For a network around a fixed point, the linear-response transfer function of each population (the Richardson single-pole form [3]) is

$$H_i(\omega) = g_i / (1 + i\omega\tau_m), \quad (3)$$

where g_i is the neuron’s gain (how strongly its rate responds to input) and τ_m its time constant. The numerator is the gain; the denominator is the low-pass roll-off. Closing the loop over a network with connectivity J , the network’s natural frequencies are the roots of $\det(I - H(\omega)J) = 0$ — classical control theory, unchanged.

Plain-language reading. $|H(\omega)|$ plotted against ω is a **Bode plot**. Reading it is reading how nimble the network is: a wide plot means it tracks fast input, a narrow one means it is sluggish. The **phase** of $H(\omega)$ says how delayed the response is, and a loop whose phase reaches a half-turn while its gain still exceeds one will oscillate — the Barkhausen criterion of electronics, which reappears literally in Section 2.4.

What it costs. Same as rung 1 — it is the same linearization, read in the frequency domain instead of the time domain. The extra information is dynamic: ringing frequencies, bandwidths, stability margins.

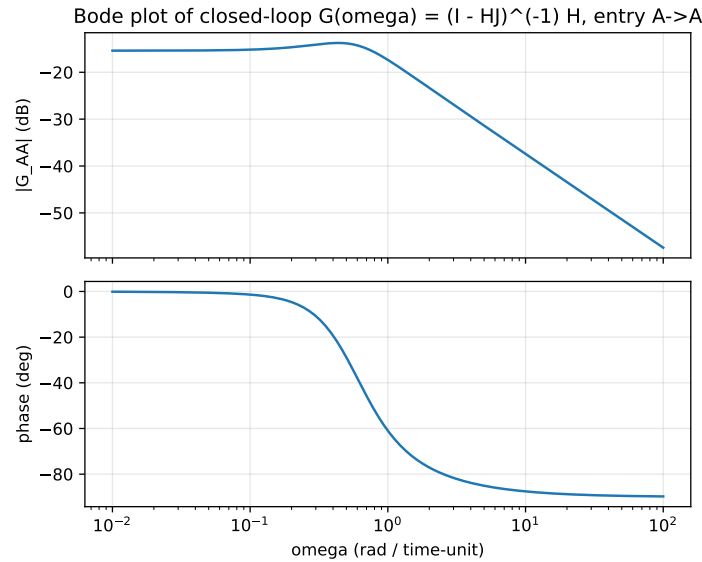


Figure 1: A Bode plot (rung 2): the closed-loop frequency response of the negative-loop archetype. The magnitude falls off as the driving frequency rises — the network is a low-pass filter — with a small resonant rise near its natural ringing rate.

Where it appears. Transfer functions diagnose **oscillation** (Section 3.2). They succeed at saying **whether and roughly how** a negative loop rings, and they fail — structurally, unfixably — at saying **at what integer period**. That failure (Section 5, dead end 4) is one of the cleanest results in the programme.

2.4. Rung 3 — Wilson–Cowan mean-field

What this method is. Rungs 1–2 quietly replaced the hard spike threshold with a smooth curve. Rung 3 says **where that smooth curve comes from**. The trick is a change of scale. Instead of one neuron with one hard threshold, consider a **population** of many similar neurons whose thresholds are slightly spread out. Ask not “did this neuron spike” but “what fraction of the population spiked”. Because the thresholds are spread, that fraction is a **smooth** function of the input — the reset discontinuity of any single neuron is averaged away by the crowd. The population’s firing rate then obeys a smooth differential equation, the **Wilson–Cowan equation** [4], and a smooth equation is exactly what bifurcation theory needs.

Replace each node of an archetype with a population of N LI&F neurons whose firing thresholds are drawn from a distribution. The fraction firing at input x is the fraction whose threshold is below x — the cumulative distribution, a smooth S-shaped **gain function** f . The population rate ρ_i then relaxes according to

$$\tau \dot{\rho}_i = -\rho_i + f\left(\sum_j W_{ij}\rho_j + I_i\right), \quad (4)$$

which reads: each population decays toward the rate its current total input calls for. This is a smooth ODE. Linearizing it (rung 1, now on solid ground) and locating where an eigenvalue crosses zero gives the **bifurcation curves** — the analytic boundaries of behaviour.

Plain-language reading. The single most useful consequence: mutual inhibition between two populations is a feedback loop whose **loop gain** is the product $w_{12}w_{21}g_1g_2$ of the two weights and the two gains. The winner-take-all transition happens exactly when that product crosses 1 — the **Barkhausen criterion** again. Because perturbations **multiply** around a loop, the boundary is a curve of constant **product** (a hyperbola), not a straight line. That is why the winner-take-all region has the shape it has, and the shape falls out of two lines of algebra.

What it costs. The same $O(n^3)$ linear algebra, plus a one-time symbolic derivation per archetype. The gain is qualitative completeness: pitchfork, Hopf, and saddle-node bifurcations are all in reach, and the FCS archetype taxonomy (contralateral, negative loop, series, parallel, positive loop) maps cleanly onto them.

Where it appears. The population thread (Section 7, thread 3) is rung 3. Its bifurcation curves are **exact** — machine-precision agreement with simulation — which is the strongest positive result in the compendium (Section 3.1). Its honest limit is that the smooth average has, by construction, discarded the reset; Section 5 dead end 5 is exactly the price.

2.5. Rung 4 — The Siegert closed form

What this method is. Rung 3 still needs a gain function f , and so far f was a hand-tuned S-curve with two free knobs. Rung 4 removes the knobs. Siegert’s 1951 formula [5] derives the gain function **from the neuron’s own physics**: if a neuron integrates noisy input and fires on threshold crossing, its firing rate is a specific closed-form expression in just the **mean** and the **variance** of its input. Nothing is fitted to behaviour — every quantity in the formula is a measurable property of the LI&F neuron.

The Siegert firing rate is

$$\nu = \Phi(\mu, \sigma) = \frac{1}{\tau_{\text{ref}} + \tau_m \sqrt{\pi} \int_{\frac{V_r - \mu}{\sigma}}^{\frac{V_{\text{th}} - \mu}{\sigma}} \text{erfcx}(-u) du}, \quad (5)$$

where μ and σ are the mean and standard deviation of the input and erfcx is a standard special function. The formula looks heavy; the reading is light: it is the smooth gain curve of rung 3, but now its shape is **dictated** by the neuron rather than chosen. The reset enters honestly, as the lower limit of the integral (the potential resets to V_r). A network’s resting state is then found by solving the **self-consistency** condition $\nu_i^* = \Phi(\mu_i(\nu^*), \sigma_i(\nu^*))$ — each population’s rate must be consistent with the input the other rates produce.

Plain-language reading. Rung 4 buys **physical honesty**. Where rung 3’s S-curve could be accused of being chosen to fit, rung 4’s curve is a prediction. Concretely, enumerating the self-consistent fixed points of Equation 5 tells you whether an archetype is **bistable** (two stable rates — a winner and a loser exist) or **monostable** (one rate — no winner), with no tuning. To make the FCS neuron’s input genuinely noisy enough for the diffusion picture to hold, the threads inject controlled randomness — small threshold jitter and random input thinning — which is a modelling choice, not a fit.

What it costs. A one-time calibration of four physical constants against the neuron’s measured input-output curve, then root-finding — still cheap.

Where it appears. The closed-form thread and its winner-take-all sibling (Section 7, threads 4–5) are rung 4. Siegert beats the hand-tuned curve and yields the 99.6%-recall **envelope** of the FCS winner-take-all region (Section 3.1). Its limit is that it is **static** — a resting-rate theory with no notion of time — and that its envelope is exactly an envelope: it over-cludes the staircase.

2.6. Rung 5 — Quasi-renewal mesoscopies

What this method is. Mean-field theory (rungs 3–4) assumes **infinitely many** neurons, so its averages are exact only in that limit. A real, finite population is noisy: the fraction firing fluctuates from tick to tick, and the size of that fluctuation scales as $1/\sqrt{N}$ for N neurons. The **quasi-renewal** equations [6] add this finite-size noise back, and — more importantly — track the population’s **age distribution**: how long ago each neuron last fired. Tracking age is how the reset gets back in: a neuron that just fired is young and cannot fire again immediately. Rung 5 is the first rung that **carries the reset**, at the price of being a stochastic rather than a deterministic description.

The mesoscopic update is the single-integral form

$$A(t) = \sum_k m_k(t-1) h(k; \mu(t)) + \sqrt{A(t)/N} \xi(t), \quad (6)$$

where m_k is the fraction of the population that last fired k ticks ago, h is the hazard (the Siegert rate, gated by how long since the last spike), and ξ is unit noise. The first term is “of the neurons eligible to fire, this fraction does”; the second is the $1/\sqrt{N}$ finite-size jitter. As $N \rightarrow \infty$ the noise vanishes and Equation 6 collapses back to the mean-field rate equation — rung 5 **contains** rungs 3–4 as its large- N limit.

Plain-language reading. Two things rung 5 recovers that rung 4 cannot. First, **time**: because it tracks age and reset explicitly, it produces an actual oscillation period, and — remarkably — the **correct** one (Section 3.2). Second, **finite-size effects**: at small N the noise blurs sharp boundaries, and that blur is sometimes exactly the correction needed to better match the discrete oracle. The reset is back in the description; what is still missing is that the description remains a **rate**, so it cannot produce a **binary** spike train.

What it costs. A stochastic simulation, swept over population size N — more expensive than rungs 1–4, still far below the model checker.

Where it appears. Quasi-renewal is the finite-size rung in every closed-form thread (Section 7, threads 4–7). It recovers the oscillation period exactly (Section 3.2) and partially dissolves the winner-take-all staircase (Section 3.1); its honest ceiling is the ≈ 0.70 Jaccard floor and the inability to produce the binary waveform.

2.7. Rung 6 — The discrete FCS oracle

What this method is. The bottom rung is the system itself. The FCS LI&F neuron, simulated or model-checked **exactly**: every integer tick, every threshold test, every reset computed with no averaging and no linearization. This is rung 6. It answers any question about the network — exact spike trains, exact periods, bit-exact verdicts — and it is what the prior CogSpike paper [1] and the FCS programme [2] use. Its cost is the state-space explosion: exhaustive verification is exponential in neuron count. Rung 6 is the reference against which rungs 1–5 are measured, and the cost of rung 6 is the reason rungs 1–5 exist.

There is nothing to approximate at rung 6, so there are no equations to introduce — only the FCS LI&F definition already given in Section 2.1, run forward tick by tick (the deterministic **oracle** used as ground truth in every thread) or compiled to Lustre and discharged to the Kind2 model checker (the verification path of [2]).

Plain-language reading. Rung 6 is the only rung that sees the binary 1100 spike pattern, the exact integer period, and the deterministic lock-in — because rung 6 is the only rung that did not throw the reset away. Everything the continuous ladder **cannot** do is, definitionally, what rung 6 is **for**. The compendium’s recurring phrase “the positive content of formal verification” means precisely the set of questions that need rung 6.

2.8. The ladder as a whole, and the diagnostic principle

Rung	Method	What it computes	Cost
1	Linearization / eigenvalues	stability, convergence rate, reachability of attractors	$O(n^3)$
2	Transfer functions $H(\omega)$	frequency response, ringing direction, stability margins	$O(n^3)$
3	Wilson–Cowan mean-field	bifurcation curves: winner-take-all and oscillation boundaries	$O(n^3)$ + symbolic
4	Siegert closed form	physically-grounded fixed points; the winner-take-all envelope	calibration + root-finding
5	Quasi-renewal mesoscopes	finite-size corrections; the oscillation period	stochastic sim, swept in N
6	Discrete FCS oracle	everything, bit-exact: spike trains, integer periods, verdicts	exponential

Table 2: **The methodology ladder.** Fidelity increases and tractability decreases down the rungs. Each rung restores one more piece of the spike physics that the rung above discarded; the boundary between rungs 1–5 and rung 6 is the spike-reset rule.

Climbing from rung 1 to rung 6, each step puts back one piece of the spike physics. Rung 1 ignores the reset entirely. Rung 3 averages it into a smooth curve. Rung 5 tracks it through the age distribution. Rung 6 computes it exactly. The single thing the rungs disagree about is the reset — and that is the diagnostic principle, now stated in full:

The diagnostic principle. The spike-reset rule is the essential nonlinearity that continuous analysis erases. Whether it appears in the formal property being verified determines whether continuous methods apply. Properties about the existence and stability of attractors — reachability, bifurcation, oscillatory-vs-not classification — do not depend on the exact reset schedule, and rungs 1–5 reach them. Properties about the exact integer schedule of spike emissions — deterministic bit-exact trajectories, specific integer periods, binary waveforms, deterministic lock-in — are the reset schedule, and only rung 6 reaches them.

This principle is not a slogan attached after the fact. It was rediscovered independently in four threads, on three mathematical frameworks (eigenvalues, transfer functions, quasi-renewal) and two phenomena (winner-take-all, oscillation). Section 3 is the evidence; Section 4 is the ledger.

3. The three phenomena

Section 2 introduced the six methods. This section applies them. The threads studied three phenomena of FCS LI&F archetypes — **winner-take-all**, **oscillation**, and **reachability** — and each subsection below takes one phenomenon and walks it down the ladder, so the reader sees the same target hit by every method in turn. The thread-level numbers live here; the per-thread chronology and scorecards are in Section 7 and Section 8.

3.1. Winner-take-all

The phenomenon. In the **contralateral inhibition** archetype, two neurons each receive constant external drive and each inhibit the other. **Winner-take-all** (WTA) is the outcome where one neuron fires steadily and the other falls silent. FCS verifies this as Property 7, and its Figure 10 plots, over a 40×40 grid of the two inhibitory weights (w_{12}, w_{21}), whether WTA stabilises within four ticks.

The discrete ground truth (rung 6). Reproducing FCS’s oracle gives $1014/1600 = 63.4\%$ WTA-stable (“blue”) cells. The striking structure is **three diagonal blocks of red** in a sea of blue: bands of weights where, instead of a winner, both neurons fall into a **synchronous lock** — they fire in identical lockstep and never break symmetry. Classifying the red cells by the period of that lock gives a clean partition: period-2 lock at weak weights (144 cells), period-3 at medium weights (361 cells), period-4 at strong weights (81 cells). These three blocks are the **staircase**, and they are the recurring antagonist of this section: every continuous method must somehow account for them, and none fully does.

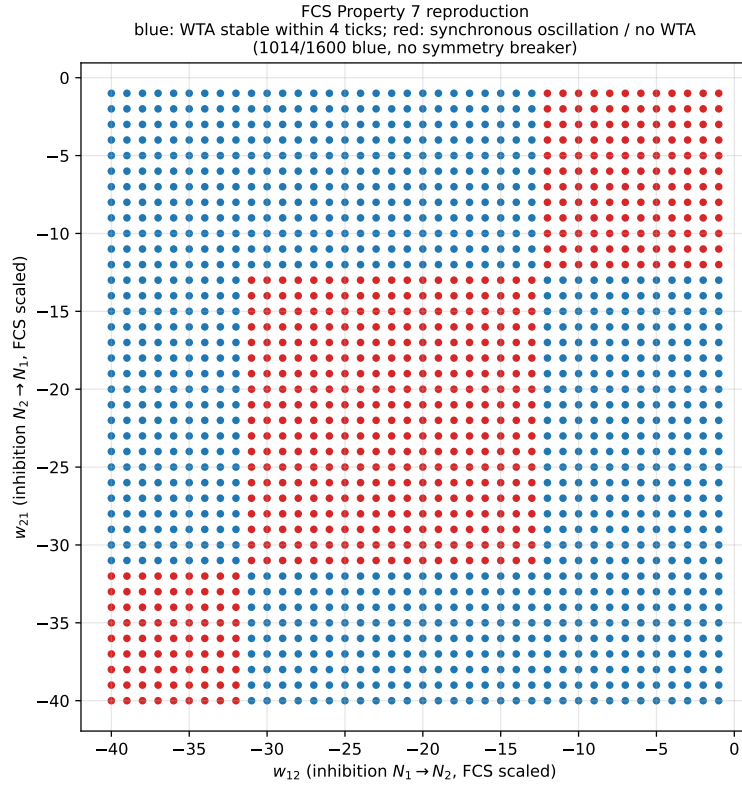


Figure 2: The discrete winner-take-all ground truth (rung 6): the FCS Property 7 / Figure 10 grid, our reproduction. 63.4% of cells are WTA-stable (blue); the rest fall into the three diagonal **staircase** blocks of synchronous lock (red). Every continuous method below is judged against this map.

3.1.1. Rung 1 on WTA: a two-regime split

The first honest test of the eigenvalue method was: does the spectral radius, or the eigenvalue gap, of the linearized contralateral system predict the blue region of Figure 2? The answer split cleanly in two, along a semantic line, and the split is the foundational thread’s central result.

It depends on **which question** “winner-take-all holds here” is asking. Two formalisations coincide for smooth systems but diverge for a bit-exact discrete one:

- **Deterministic semantics:** starting from the zero initial state, does the one trajectory the simulator follows reach WTA?
- **Reachability semantics:** does **some** trajectory, from **some** small perturbation of the initial state, reach WTA?

Kind2 model checking searches over reachable states — it verifies the **second**. A physical network from fixed initial conditions runs the **first**.

Finding. Under **deterministic** semantics, WTA is **combinatorial**, not spectral. No eigenvalue quantity beats 64–69% classification accuracy against the 63.4% blue baseline. The eigenvalue gap of the raw weight matrix is in fact **identically zero** for any 2×2 zero-diagonal matrix (a structural feature of mutual inhibition), and the linearized spectral radius is symmetric under swapping the two weights, hence blind to the asymmetry that decides the winner. What **does** decide it is an integer comparison: the simulator breaks symmetry at tick 2 via the predicate $(|w_{12}| > 12) \text{ xor } (|w_{21}| > 12)$, and the winner is the neuron receiving weaker inhibition. The plain rule $\|w_{12}\| - \|w_{21}\| > 7$ classifies the deterministic map at 83.4% — better than any spectral quantity — and the sign of $|w_{12}| - |w_{21}|$ names the winner with 100% accuracy.

The same eigenvalue method, asked the **reachability** question, succeeds:

Finding. Under **reachability** semantics — the semantics Kind2 actually verifies — WTA is spectral. The reachability ground truth has 97.8% blue cells (a 6×6 corner of weak weights is the only unreachable region), and the scalar linearization’s spectral radius separates the classes cleanly: unreachable cells have $\rho(A) \in [0.518, 0.598]$, reachable cells $\rho(A) \in [0.544, 0.929]$. The predicate $\rho(A) < 0.544$ is a **sound** certificate of non-reachability — every cell it flags is genuinely non-reachable — and overall classification reaches 98.5%.

This two-regime split Figure 3 is the cleanest illustration of the diagnostic principle in the whole programme. The reset is the mechanism that **chooses** the winner under deterministic semantics, so deterministic WTA is out of reach of linearization; under reachability semantics the reset’s effect is averaged over perturbations, so reachability WTA is in reach.

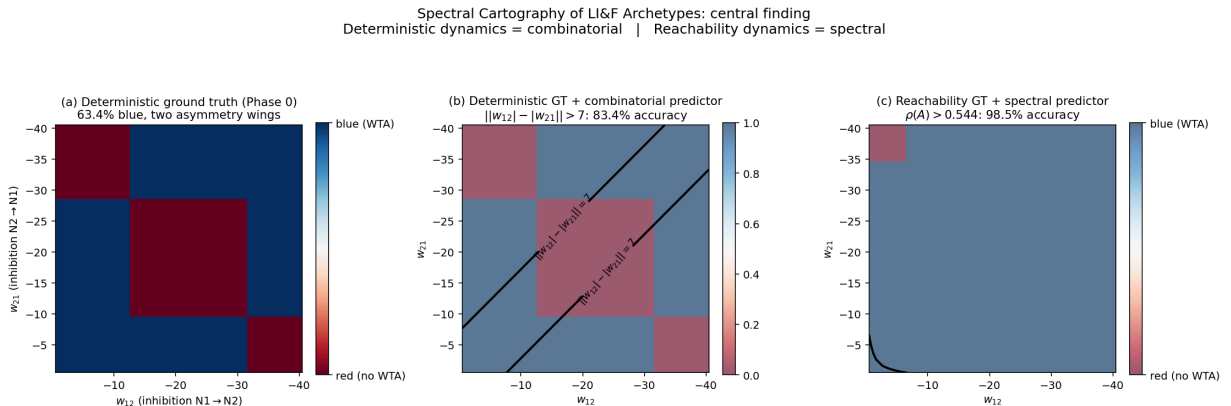


Figure 3: **The two-regime split.** (a) Deterministic ground truth, 63.4% blue. (b) The combinatorial predictor $\|w_{12}\| - \|w_{21}\| > 7$ traces the deterministic boundary at 83.4% — no spectral quantity reaches this. (c) Reachability ground truth, 97.8% blue; the spectral-radius contour $\rho(A) = 0.544$ (black) bounds the unreachable region at 98.5%. Panels (b) and (c) are the finding: the deterministic map is combinatorial, the reachability map is spectral.

3.1.2. Rung 3 on WTA: the bifurcation curve

Lifting the contralateral archetype to a Wilson–Cowan population pair makes the WTA boundary a **bifurcation curve** that can be written down. Linearizing Equation 4 and decomposing into sum and difference modes shows the symmetric (“no winner”) state loses stability exactly when the loop gain $w_{12}w_{21}g_1g_2$ crosses 1 — a **pitchfork** bifurcation. The dominant eigenvalue of the mean-field Jacobian then classifies WTA on a 50×50 grid at 99.96% accuracy, and the pitchfork curve itself, traced symbolically from the loop-gain-unity condition, agrees with the numerical bifurcation trace to machine precision.

Finding. In the mean-field description, winner-take-all is an **exactly solvable** bifurcation. The boundary is the loop-gain-unity hyperbola $w_{12}w_{21}g_1g_2 = 1$; a single eigenvalue predicts which side of it a cell is on at 99.96%. This is the strongest positive result in the compendium: it is the “shape invisible to the discrete lens” the programme set out to find. The whole FCS archetype taxonomy follows the same pattern — contralateral inhibition gives a pitchfork, the negative loop a Hopf, the positive loop a saddle-node, and the sign of the loop’s weight product selects which (Thomas’s rule).

But the mean-field WTA boundary is not the **discrete** WTA boundary. Validated against the LI&F simulator on the same grid, the smooth pitchfork hyperbola agrees only at the symmetric corner. The discrete bistable region is a pair of **rectangular strips** (a winner emerges once **either** weight exceeds ≈ 7), because once either neuron fires first, its inhibition can lock the other below threshold — a **spike-timing lock-in** that the smooth average has no term for.

Partial result. The Wilson–Cowan pitchfork is a **lower bound** on the discrete winner-take-all region, not its envelope. The smooth average captures one mechanism of bistability — the symmetric state losing stability — and misses a second, **spike-timing lock-in**, which is intrinsically discrete. The two are complementary mechanisms at two scales, not a right and a wrong answer; but a reader who needs the discrete boundary cannot get it from rung 3.

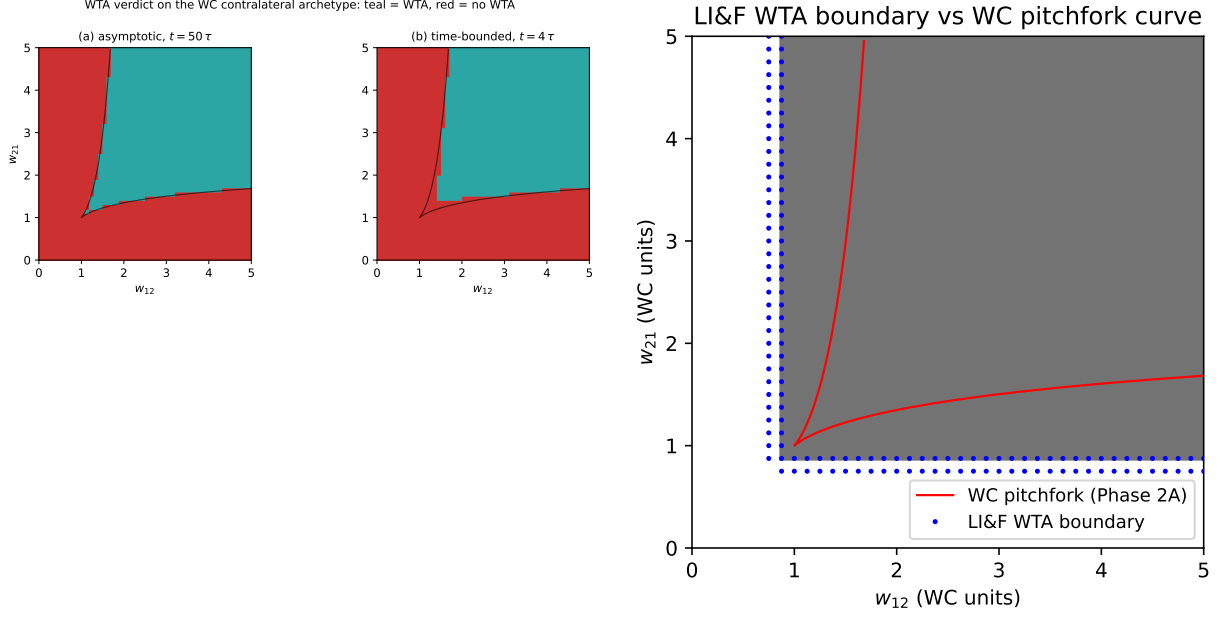


Figure 4: Rung 3 on WTA. **Left:** the mean-field winner-take-all region with the symbolic pitchfork curve overlaid; the dominant eigenvalue classifies it at 99.96%. **Right:** the discrete LI&F bistable region (grey) against the same pitchfork curve (red) — they agree at the symmetric corner and diverge in the arms, where spike-timing lock-in, invisible to the smooth average, dominates.

3.1.3. Rung 4 on WTA: the Siegert envelope

The closed-form thread replaces rung 3’s hand-tuned gain curve with the Siegert formula Equation 5 and re-reads the FCS staircase grid. Enumerating the self-consistent fixed points and marking a cell WTA-capable when a bistable pair of rates exists gives a map that relates to the discrete oracle in a specific, useful way:

Finding. The Siegert fixed-point map is a high-recall **envelope** of the FCS winner-take-all region: it recovers 99.6% of the FCS-blue cells. Its errors are almost all one-sided — it labels the staircase blocks blue when FCS calls them red. The reading is that rate-equation bistability is **necessary** for WTA but not **sufficient**: the staircase cells are bistable in the rate description, but the integer-tick dynamics never commit to either branch. Because the errors are one-sided, the envelope is a sound **pre-filter**: any cell **outside** it is guaranteed non-WTA and need not be model-checked — a 7.4% reduction in the Kind2 workload on this grid.

The Siegert curve also beats the hand-tuned curve on its own terms: calibrated against the neuron’s measured input–output curve, it agrees with the discrete WTA region at Jaccard 0.843 versus the hand-tuned curve’s 0.796. Rung 4’s limit is structural — it is a **static** theory, so its overall Jaccard with the staircase-bearing FCS oracle sits at 0.680, the staircase being exactly the cells it over-cludes.

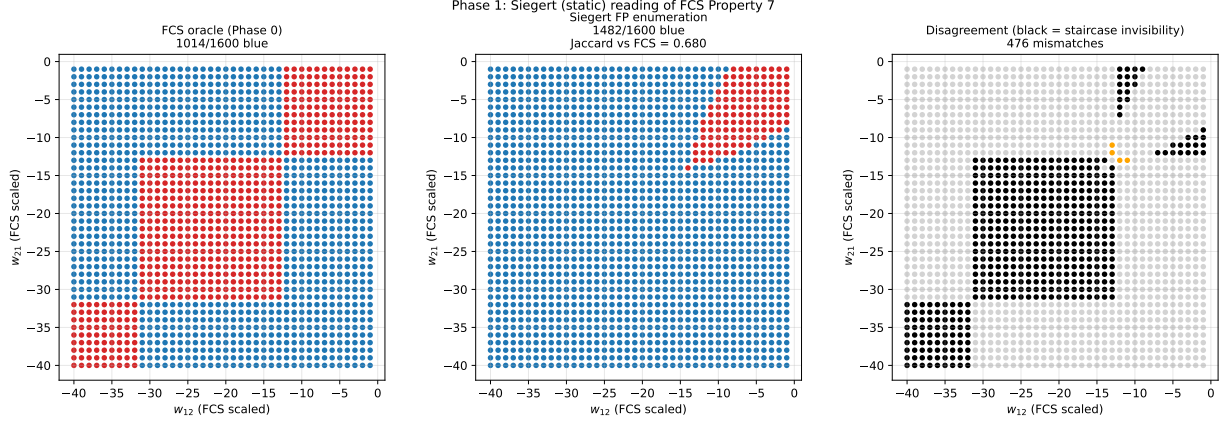


Figure 5: Rung 4 on WTA. Left: the FCS oracle. Centre: the Siebert fixed-point map. Right: the disagreement — black cells are Siebert-blue, FCS-red, i.e. exactly the diagonal staircase. Siebert recovers 99.6% of FCS-blue cells but adds the staircase as one-sided false positives, which is what makes it a sound pre-filter envelope.

3.1.4. Rung 2 on WTA: a negative result

It is natural to hope that the staircase — where the discrete system fails to **commit** within four ticks — is a slow-dynamics effect, so that a transfer-function latency gate (“is the slowest mode fast enough?”) would separate the staircase from the genuine WTA cells. It does not.

Negative result. The transfer-function latency gate is **orthogonal** to the staircase. Gating cells by $|\text{Re}(\lambda_{\text{dom}})| > 1/4$ changes the Jaccard agreement with FCS by -0.003 — statistically nothing. The eigenvalue distributions of staircase and non-staircase cells overlap almost completely: the staircase cells are **not** slow-decay cells. They have well-separated fixed points and respectable decay rates. Their redness is integer-tick determinism, which has no signature in any linear timescale.

This is a useful negative: it rules out the most plausible rung-2 rescue and sharpens the diagnosis. The staircase is not “slow”; it is **discrete**.

3.1.5. Rung 5 on WTA: partial dissolution

Quasi-renewal adds finite-size noise. Since the staircase is a fragile synchronous lock, noise should occasionally knock a locked cell off its orbit and let it commit to a winner — partially dissolving the staircase and improving agreement with FCS. It does, partially.

Partial result. Finite-size noise partially dissolves the staircase. Agreement with the FCS oracle is best at **small** population size — Jaccard 0.701 at $N = 50$, rising to a peak of 0.717 at $N \approx 20$ — because more noise dissolves more lock. But the agreement is **unimodal** in N , not monotone: below $N \approx 20$ the same noise also destabilises genuine WTA cells, and the two effects trade off. The improvement over the static Siebert envelope is real but small ($+0.04$ Jaccard), and it is concentrated on the period-3 and period-4 blocks. As $N \rightarrow \infty$ the noise vanishes and quasi-renewal converges back to the Siebert mean-field (Jaccard 0.963), confirming it sits correctly above rung 4 on the ladder.

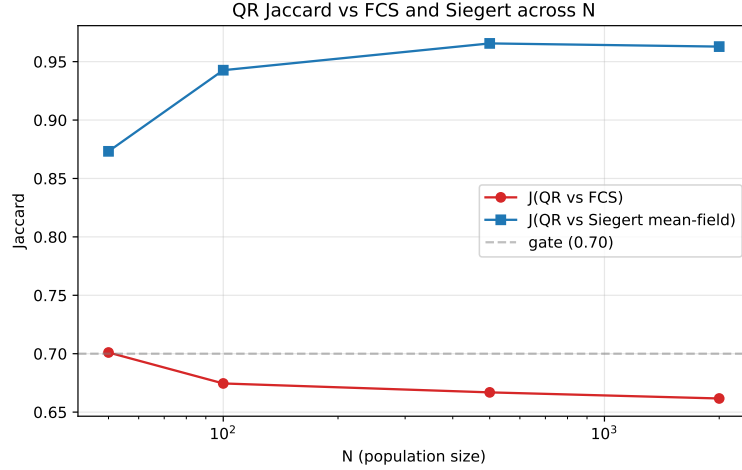


Figure 6: Rung 5 on the winner-take-all staircase: agreement with the discrete oracle against population size N . Agreement is **unimodal** — best near $N \approx 20$, where finite-size noise dissolves the most synchronous lock before it begins to destabilise genuine winner-take-all cells — and converges back to the static Siegert envelope as $N \rightarrow \infty$.

3.1.6. The Jaccard floor, and the multi-neuron inverse staircase

Across rungs 1–5, no rate-equation method exceeds a Jaccard agreement of **about 0.70** with the discrete FCS winner-take-all oracle. This ceiling is structural — it is not closed by more noise, a better gain curve, or a finer linearization.

The ≈ 0.70 Jaccard floor is the positive content of formal verification. The residual ≈ 0.30 disagreement between every rate-equation method and the discrete oracle is the integer-tick synchronous lock — the staircase — and it lies on a genuine feature of the dynamics (deterministic spike-timing lock with phase memory) that no rate-and-hazard theory can express. It is, precisely, what only rung 6 can verify.

Extending the contralateral archetype from two neurons to $N > 2$ (uniform all-to-all inhibition) does not just preserve this gap — it adds a **second**, opposite failure. The two-neuron Siegert envelope had 99.6% recall; at $N > 2$ recall **collapses** with N (from 0.90 at $N = 2$ to 0.00 at $N = 10$). And at $N = 10$ the discrete oracle shows clean WTA at specific integer weights where Siegert finds **no asymmetric fixed point at all**.

Negative result. The high-recall Siegert envelope is a two-neuron luxury. At $N > 2$ a new phenomenon appears — the **inverse staircase**: the discrete FCS dynamics produce clean winner-take-all **below** the smooth-rate bifurcation threshold, in cells where the rate equations admit only the symmetric no-winner fixed point. At $N = 2$ the continuous lens **over**-predicts WTA (the staircase); at $N = 10$ it **under**-predicts it (the inverse staircase). The continuous lens fails in both directions, and neither failure is closed by finite-size noise.

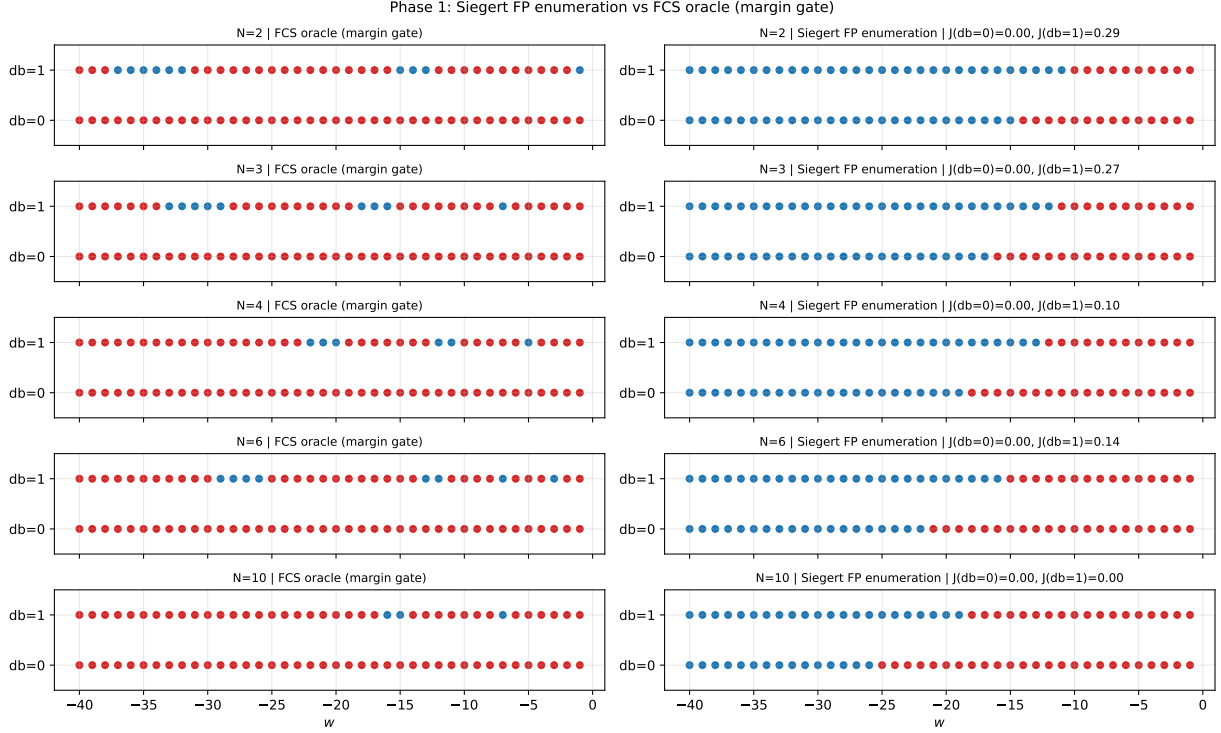


Figure 7: The multi-neuron inverse staircase. Siegert fixed-point enumeration (right of each pair) versus the FCS oracle (left), per neuron count N . At large N the discrete oracle shows winner-take-all where Siegert finds only the symmetric fixed point — the continuous lens now **under-predicts**, the mirror image of the two-neuron staircase.

3.2. Oscillation

The phenomenon. In the **negative loop** archetype, an activator A excites an inhibitor I , which inhibits A back. Under constant drive the activator settles into a periodic spike pattern. FCS verifies this as Property 5: with the default weights the activator fires the **period-4** pattern **1100** repeating, and the inhibitor echoes it one tick later. Reproducing the oracle over the weight grid, 27.8% of cells produce that strict **1100** pattern and 59.1% produce some regular oscillation.

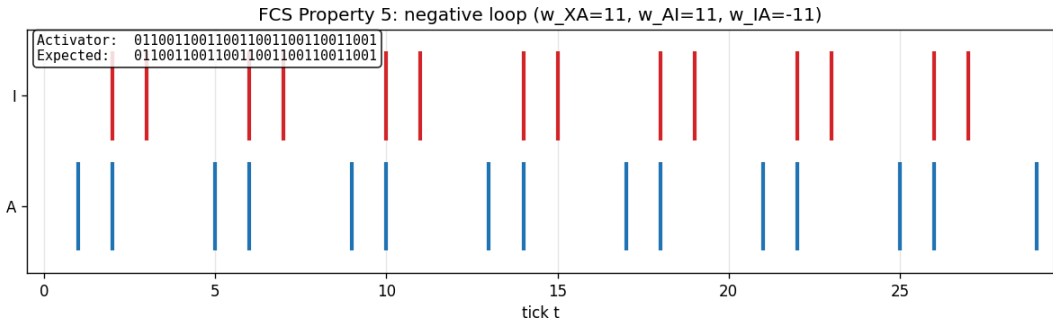


Figure 8: The discrete oscillation ground truth (rung 6): activator (blue) and inhibitor (red) spike rasters of the negative loop at the default weights. The activator fires the exact FCS Property 5 pattern **1100** repeating, period 4; the inhibitor is the activator delayed one tick.

Oscillation exposes the ladder differently from winner-take-all, because here the **mean-field theory cannot even produce the phenomenon**.

3.2.1. Rungs 3–4 on oscillation: a stable spiral everywhere

For the negative loop the rate-equation Jacobian has eigenvalues $\lambda_{\pm} = (-1 \pm \sqrt{g_A g_I w_{AI} w_{IA}}) / \tau_m$. Because the loop is inhibitory, the term under the square root is negative, so λ is a complex pair with **negative** real part $-1/\tau_m$ — a **stable spiral**. The rate equations describe a system that **rings and then settles**, never one that oscillates forever.

Negative result. Mean-field theory cannot create the FCS oscillation. The negative loop’s rate-equation fixed point is a **stable spiral for every weight choice** — the real part of the eigenvalue is pinned negative. Property 5’s sustained oscillation is therefore **not** a Hopf bifurcation of the rate equations; it lies strictly beyond mean-field reach. The Siegert “spiral envelope” (cells whose fixed point has a complex eigenvalue) covers 90% of the grid but agrees with the strict **1100** region at Jaccard only 0.31 — it marks where ringing is **possible**, not where the integer-tick limit cycle locks.

This is a sharper limit than rung 3’s winner-take-all limit. There, mean-field got a **lower bound**; here it gets the wrong **type** of object — a decaying spiral instead of a sustained cycle.

3.2.2. Rung 2 on oscillation: the right direction, the wrong period

The transfer function turns the spiral’s imaginary part into a predicted ringing period $T_{\text{pred}} = 2\pi / |\text{Im}(\lambda)|$. This gets the **qualitative** behaviour right — it correctly says the loop rings, and that the ringing rate grows with the inhibition strength. But the **period** is wrong, and wrong in a very particular way.

Negative result. The transfer-function period prediction is too long by a **structural factor of about four**. Across the entire FCS period-4 region the predicted period lands on the line $T_{\text{pred}} = 4T_{\text{FCS}}$ — a constant multiple, not scatter. The obvious fix — the calibration is off, so recalibrate the time constant — was tried (followup A of the negative-loop thread) and **falsified**: refitting the time constant against the neuron’s dynamic response made the gap **wider**, not narrower. The reason is structural: at the oscillation frequency the neuron’s linear response magnitude is ≈ 0.075 — essentially flat. The closed-loop oscillation lives deep in the rolled-off regime of the single-neuron filter, where no single-pole model at any time constant has a resonance to offer. The factor of four is not a calibration error; it is the linear-response framework reaching past its domain.

This is the cleanest **epistemic** result in the programme: a tempting explanation (recalibration) was stated as a hypothesis, tested, and killed with a mechanism. The factor-of-four is reclassified from “a bug to fix” to “a provable limit”: no single-pole transfer function can predict the period.

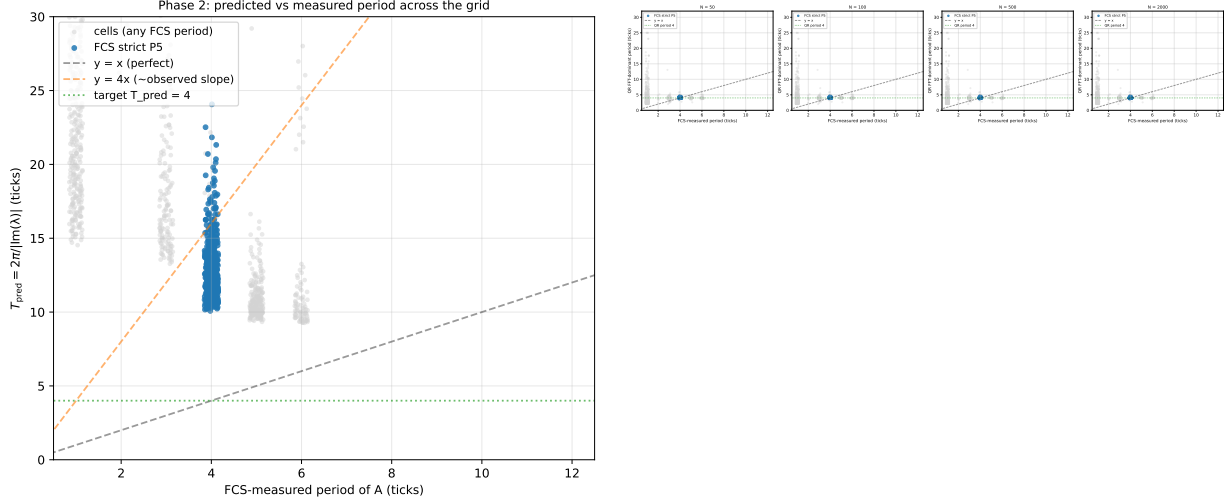


Figure 9: Oscillation period: rung 2 versus rung 5. **Left:** the transfer-function prediction lands on the line $T_{\text{pred}} = 4T_{\text{FCS}}$ (orange) — a structural factor-of-four overshoot. **Right:** the quasi-renewal prediction lands on the diagonal $T = T_{\text{FCS}}$ — the correct integer-tick period.

3.2.3. Rung 5 on oscillation: the period recovered

Quasi-renewal, which carries the reset through the age distribution, succeeds exactly where the transfer function fails.

Finding. Quasi-renewal recovers the oscillation period. Using the **same** calibration the transfer function used, it predicts a period of 4.05 ticks at the default cell — essentially exact agreement with the FCS period of 4 — and across the grid its predictions land on the correct diagonal rather than the factor-of-four line. The difference from rung 2 is not the parameters; it is the **mechanism**: the age-distribution stepper represents spike-and-reset explicitly, and “reset” is exactly what a single-pole filter cannot express. The sustained oscillation itself is finite-size: its amplitude decays from 0.20 at $N = 50$ to 0.05 at $N = 2000$, confirming that the cycle is noise-driven — the deterministic mean field still only spirals inward.

What rung 5 still cannot deliver is the **binary waveform**. Its output is a smooth rate, so even when its period is exact its match to the binary 1100 template is only ≈ 0.28 . A companion experiment confirmed this is structural: a deterministic single-neuron renewal predictor also recovers the period (4.00) exactly but, being a rate, cannot sharpen into a binary spike train. The waveform is rung 6’s alone.

A three-neuron negative loop closes the section cleanly: the period scales as $2(n+1)$ in the number of delayer stages n , quasi-renewal tracks that scaling (6.38 predicted versus 6 exact for the three-neuron loop), and the transfer function stays structurally wrong. The ladder’s verdict on oscillation generalises.

Experiment C: 3-neuron negative loop ($A \rightarrow D \rightarrow I \rightarrow A$) closed-form lenses

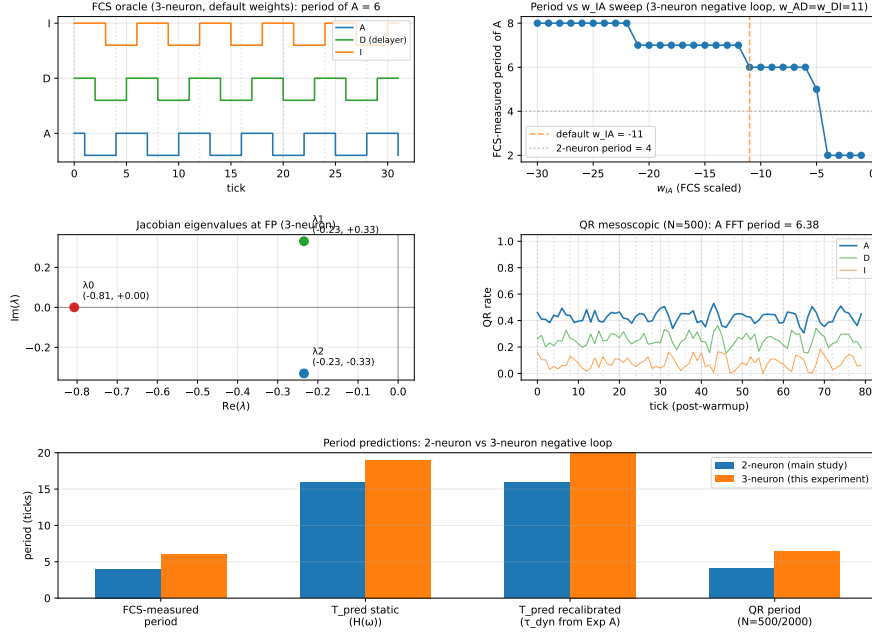


Figure 10: The three-neuron negative loop. The discrete period is 6 ticks — the rule $2(n + 1)$ for n delayer stages — and quasi-renewal tracks it (6.38 predicted); the transfer function remains structurally wrong. The ladder’s verdict on oscillation is not a two-neuron artefact.

3.3. Reachability

The phenomenon. **Reachability** asks: can the system reach a target behaviour from **some** admissible initial state? It is not a separate archetype — it is the question a model checker actually answers, since Kind2 searches over reachable states. It is treated as its own phenomenon here because it is the one place the continuous lens delivers a clean, **sound** positive result, and because it pins down exactly which formalisation of a property the continuous methods match.

The reachability story was told in Section 3.1.1: under reachability semantics, winner-take-all **is** spectral. The scalar linearization’s spectral radius separates reachable from non-reachable contralateral cells at 98.5%, and the predicate $\rho(A) < 0.544$ is a **sound** non-reachability certificate. Two points complete the picture.

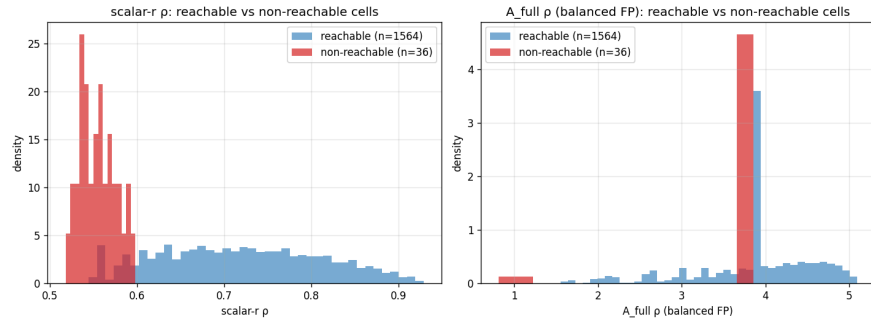


Figure 11: Why reachability is the ladder’s cleanest success: the linearized spectral radius $\rho(A)$ cleanly separates reachable from non-reachable contralateral cells. The non-reachable cells occupy a narrow low- ρ band, so $\rho(A) < 0.544$ flags them with no false positives — a **sound** certificate.

Finding. Reachability is the continuous ladder’s cleanest success, and it is sound. The predicate $\rho(A) < 0.544$ never flags a reachable cell as non-reachable, so it can be trusted as

a **pre-filter**: a cell it flags is provably not winner-take-all under the semantics Kind2 checks, and need not be model-checked at all. This composes the polynomial-cost rung 1 in front of the exponential-cost rung 6 — exactly the multi-scale workflow the ladder is built for. The saving on the contralateral grid is modest (a one-sided certificate), but the **principle** — a sound spectral certificate of a model-checking outcome — is the engineering payoff of the whole programme.

Finding. The two-regime split is a property of the **archetype class**, not of one topology. Inserting a delayer neuron into the contralateral loop (FCS Figure 11) reproduces FCS’s “contrary to expectation” asymmetric winner map — the neuron whose incoming inhibition is delayed wins more often, 1136 cells to 448 — and the spectral radius still predicts the reachability ground truth (95.8%) while missing the deterministic one. The diagnostic principle transfers unchanged.

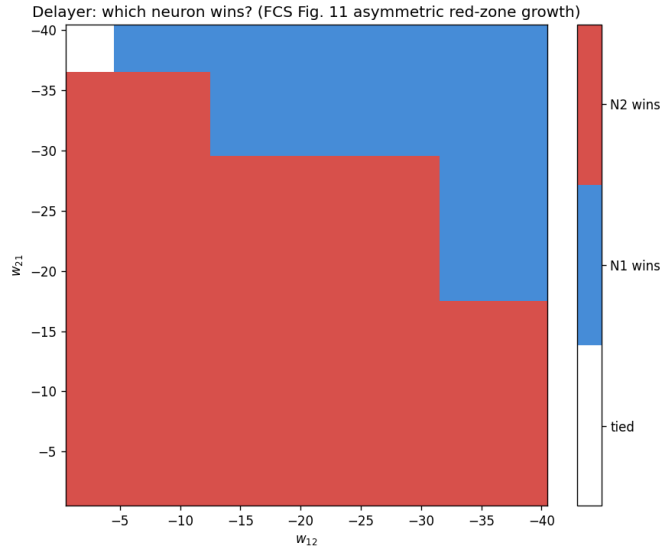


Figure 12: The delayer-augmented contralateral topology (FCS Figure 11). The neuron whose incoming inhibition is delayed wins in 1136 cells against 448 — FCS’s “contrary to expectation” asymmetry, reproduced quantitatively. The two-regime split of Section 3.1.1 holds here unchanged.

Why does reachability succeed where deterministic winner-take-all and exact oscillation fail? The diagnostic principle answers it directly. Reachability is a question about the **existence** of a trajectory to an attractor — a topological fact about the basins, which the linearized spectrum sees. The exact deterministic trajectory, and the exact period, are facts about the **reset schedule**, which it does not. Reachability is in scope because, by construction, it averages the reset over perturbations.

4. Capability and limitations matrix

Section 2 is the methods; Section 3 is the results. This section is the index that crosses them. Table 3 gives, for every rung and every phenomenon, the verdict and the one number that supports it. It is the document’s quick-reference card.

Rung	Winner-take-all	Oscillation (neg. loop)	Reachability
1 — Linearization / $\rho(A)$	fail — deterministic (64–69% vs 63.4% baseline)	fail — period indistinct, args cluster across periods 3–8	pass — $\rho(A) < 0.544$ sound; 98.5% overall
2 — Transfer function $H(\omega)$	fail — latency gate orthogonal to staircase ($\Delta J = -0.003$)	partial — ringing direction right; period $\approx 4 \times$ too long (structural)	— addressed at rung 1
3 — Wilson–Cowan mean-field	partial — gap 99.96% in mean-field; a lower bound on the discrete region	fail — fixed point is a stable spiral; no sustained cycle	pass — bifurcation-free regions certifiable
4 — Siegert closed form	partial — 99.6% -recall envelope; staircase = false positives; recall collapses for $N > 2$	fail — static; no dynamics	pass — fixed-point enumeration as pre-filter
5 — Quasi-renewal mesoscopies	partial — partial dissolution; ≈ 0.70 Jaccard floor	pass — period 4.05 exact; sustained at finite N	pass — finite-size-corrected boundary
6 — Discrete FCS oracle	pass — bit-exact staircase	pass — exact 1100, period 4	pass — the reference oracle

Table 3: **Capability matrix.** Verdict of each ladder rung on each phenomenon, with the supporting number. The pattern is the diagnostic principle made visible: rungs 1–5 pass on the “shape and existence” questions (reachability, bifurcation curves, envelopes) and fail on the “exact integer schedule” questions (the staircase, the period, the waveform); only rung 6 passes everywhere, at exponential cost.

The matrix reads diagonally. The top-left-to-bottom-right tendency is not an accident: it is the diagnostic principle. Questions about **shape and existence** — is there a winner, where is the boundary, can the target be reached — are answered by cheap rungs. Questions about the **exact integer schedule** — which staircase cell, which period, which waveform — are answered only by rung 6.

Stated as a list, here is the one hard limit each rung carries — not as an apology, but as the precise edge of its domain:

- **Rung 1** cannot reach an integer comparison. Deterministic winner-take-all is decided by a tick-2 integer test; no spectral quantity sees inside it.
- **Rung 2** cannot reach an absolute period. At the oscillation frequency the single-neuron response is flat, so no single-pole model resonates there — the factor-of-four is structural.
- **Rung 3** cannot see spike-timing lock-in. Its winner-take-all region is a provable lower bound on the discrete one; a second, discrete bistability mechanism is invisible to the smooth average.

- **Rung 4** cannot see the staircase, and its high-recall envelope is a two-neuron luxury — recall collapses, and inverts, for $N > 2$.
- **Rung 5** cannot produce a binary waveform. It recovers the period exactly but its output is a smooth rate; the **1100** pattern is beyond it.
- **Rung 6** cannot scale. It is bit-exact and answers everything, at a cost exponential in network size — which is the entire reason rungs 1–5 exist.

5. Dead ends, honestly

A compendium that reported only the successes would misrepresent the research and, worse, waste the most useful part of it. Several specific hypotheses were stated, pursued, and **failed**. They are collected here — each with what was hoped, what was tried, why it failed, and what the failure taught — because, read together, they are not a list of mistakes but a **map of the boundary**. Every one of them fails exactly where the diagnostic principle predicts; each is therefore a confirmation of the principle from its negative side.

Negative result. Dead end 1 — the eigenvalue gap predicts deterministic winner-take-all. Hoped: the eigenvalue gap of the weight matrix, or of the linearized Jacobian, classifies the FCS WTA grid. Tried: every spectral quantity, on both the scalar and the full windowed-integrator linearization. Failed: best accuracy 64–69%, barely above the 63.4% baseline; the raw-weight gap is **identically zero** for 2×2 mutual inhibition, and the Jacobian spectral radius is symmetric in the two weights and so blind to the asymmetry that picks the winner. Taught: deterministic WTA is decided by an integer comparison at tick 2, not by any continuous quantity — and this is what first forced the deterministic-versus-reachability distinction.

Negative result. Dead end 2 — a fuller linearization fixes dead end 1. Hoped: the scalar leak approximation was too crude; the full linearization that preserves the 5-tap windowed integrator would recover the signal. Tried: the $5n$ -dimensional state matrix, with three predictors. Failed: 46% accuracy at the symmetric fixed point — **worse** than the baseline — and an eigenvector-asymmetry predictor at 11%, worse than chance. Taught: the missing information is not linear. More linear detail does not help, because the quantity that decides the outcome is not in the linear dynamics at all. The plain combinatorial rule $\|w_{12}\| - \|w_{21}\| > 7$ reaches 83.4% — no linearization came close.

Negative result. Dead end 3 — pole placement designs an oscillation period. Hoped: if the spectrum predicts dynamics, invert it — place the dominant poles to realise a target integer period. Tried: solving for weights that put the poles at the target angle, in both the archetypes thread and the population thread. Failed: the dominant pole angle clusters in a narrow band across simulator periods 3 through 8, so it cannot discriminate them; inverse design missed 5 of 6 targets. (In the smooth Wilson–Cowan setting the **linear** placement is exact, but the realised nonlinear limit cycle still missed 2 of 8 near a degenerate bifurcation.) Taught: the integer period is set by the reset schedule, which the linear spectrum does not encode. Direct enumeration — a one-second brute-force sweep — subsumes pole placement outright.

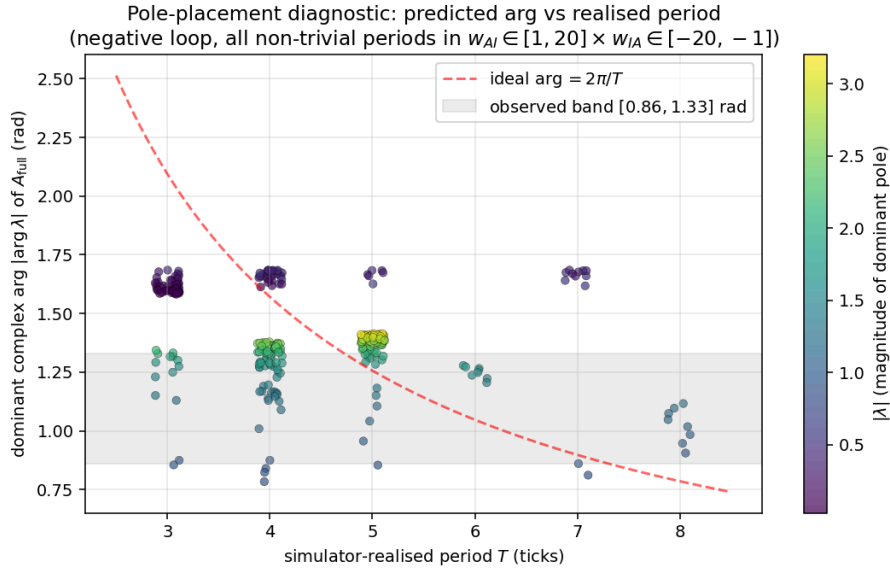


Figure 13: Dead end 3 made visible: the dominant pole angle of the linearized negative loop, against the integer period the simulator actually produces. The angle clusters in a narrow band across periods 3–8, so it cannot discriminate them — the integer period is set by the reset schedule, not the linear spectrum.

Negative result. Dead end 4 — recalibration fixes the oscillation period. Hoped: the transfer-function period is a structural $4 \times$ too long because the time constant was calibrated on static data; refit it on dynamic data and the gap closes. Tried: a Bode-style sweep of the FCS neuron’s dynamic response, then a refit. Failed: the refitted time constant made the prediction **worse** — the gap widened. The mechanism: at the oscillation frequency the neuron’s linear-response magnitude is ≈ 0.075 , essentially flat, so no single-pole model at any time constant has a resonance there. Taught: the factor-of-four is **structural**, not parametric. This converts a “method has a $4 \times$ error” into the stronger statement “**no** single-pole transfer function can predict this period, and here is why” — a negative result upgraded to an impossibility result.

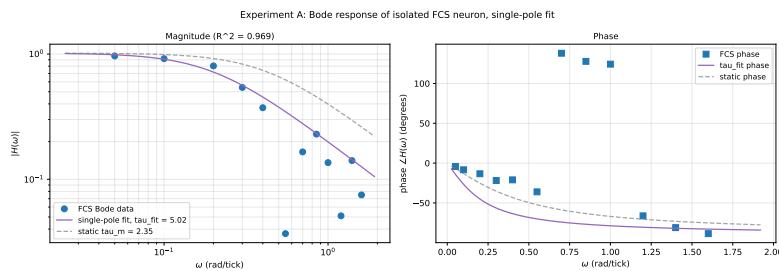


Figure 14: Why dead end 4 is structural. The measured frequency response of the FCS neuron is essentially flat (≈ 0.075) at the oscillation frequency: the closed-loop oscillation lives deep in the filter’s rolled-off regime, where no single-pole model at any time constant has a resonance to offer.

Negative result. Dead end 5 — the Wilson–Cowan pitchfork is the discrete winner-take-all boundary. Hoped: the exactly-solvable mean-field pitchfork curve is the LI&F winner-take-all boundary. Tried: cross-validating the symbolic pitchfork against the discrete simulator on the same grid, out to extreme weights. Failed: the discrete bistable

region is a pair of rectangular strips (a winner emerges once **either** weight exceeds ≈ 7); the pitchfork is a hyperbola; they coincide only at the symmetric corner. Taught: the pitchfork is a **lower bound**. The smooth average captures one bistability mechanism and misses a second, spike-timing lock-in, which is intrinsically discrete. This is the most productive dead end — it reframed a quantitative “failure” as the discovery of **two complementary mechanisms at two scales**.

Negative result. Dead end 6 — a latency gate separates the winner-take-all staircase. Hoped: FCS’s “stabilise within four ticks” gate is a speed test, so a transfer-function latency gate would carve the staircase out of the WTA region. Tried: gating cells by the slowest-mode decay rate. Failed: the gate changed agreement with FCS by -0.003 ; the eigenvalue distributions of staircase and non-staircase cells overlap almost entirely. Taught: the staircase cells are not slow — they have respectable decay rates. Their redness is integer-tick determinism, orthogonal to any linear timescale. This ruled out the most plausible rung-2 rescue and sharpened the staircase diagnosis.

Negative result. Dead end 7 — the Siegert envelope generalises beyond two neurons. Hoped: the 99.6%-recall two-neuron WTA envelope carries to $N > 2$. Tried: Siegert fixed-point enumeration for $N = 2, 3, 4, 6, 10$. Failed: recall degrades monotonically with N — 0.90 at $N = 2$ down to 0.00 at $N = 10$ — and at large N the discrete oracle shows winner-take-all where **no** asymmetric fixed point exists (the **inverse staircase**). Taught: the high-recall envelope is a small- N luxury. At larger N the continuous lens fails in the **opposite** direction, under-predicting WTA; the continuous lens is wrong on **both** sides of the discrete boundary.

Every dead end above fails exactly where Section 2.8’s diagnostic principle predicts: each one tried to extract, from a continuous description, a fact about the integer reset schedule. The seven failures, on three mathematical frameworks and two phenomena, are not seven accidents — they are seven confirmations of a single predictive principle from its negative side. Reported together, they measure the boundary of continuous analysis instead of merely bumping into it.

6. Connection to the discrete programme, and the path to publication

6.1. Back to the discrete CogSpike paper

The prior CogSpike paper [1] and this compendium are two responses to the **same** problem — the state-space explosion of exhaustive SNN verification — pulling in opposite directions. The discrete paper keeps the model exact and **shrinks the chain**: its weight-discretized quotient abstraction collapses bisimilar states, cutting a contralateral-inhibition model from 3603 to 387 states while preserving every verified winner-take-all property. This compendium keeps the cost low and **leaves the chain**: it replaces exhaustive enumeration with continuous approximation, accepting that some properties (the staircase, the period, the waveform) are lost in exchange for polynomial cost and analytic boundaries.

The two are genuinely complementary, and the same case study — contralateral inhibition, winner-take-all — runs through both, which makes the pairing concrete rather than rhetorical. The capability matrix (Table 3) is, in effect, the contract between them: it states which questions the continuous companion answers cheaply and which must still go to the discrete model checker.

The diagnostic principle is the clause that makes the contract **decidable in advance** — before running anything, one can say whether a given property is in continuous scope.

One subtlety worth recording: the discrete paper’s neuron is a **probabilistic** LI&F (threshold levels with firing probabilities, a Markov chain), while the FCS oracle used as ground truth here is the **deterministic** FCS LI&F (windowed integrator, integer threshold). They are two LI&F variants in one programme; the continuous ladder was developed and validated against the deterministic one.

6.2. Forward to a paper

This compendium is the internal, comprehensive record. A companion paper draft, `paper/continuous_lens.typ`, distils it to a publishable narrative whose thesis is the diagnostic principle: continuous analysis recovers the shape and scale of LI&F archetype behaviour, and the spike-reset rule is a decidable, a-priori boundary on which formal properties it can certify. The paper leads with the two-regime winner-take-all split as its case study, keeps the dead ends as primary evidence rather than caveats, and closes on the methodology ladder and the pre-filter workflow.

The single most publication-ready result is the **sound spectral pre-filter**: a polynomial-cost certificate ($\rho(A) < 0.544$) that provably skips model-checking cells, composed in front of the exponential model checker. It is modest in saving on the small archetypes measured here, but it is sound, and it is a concrete instance of the continuous and discrete lenses working together.

6.3. Open follow-ups

Consolidating the open directions named across the seven threads, in decreasing order of confidence:

- **Hybrid-system spectral theory.** Pole placement failed (dead ends 3–4) because the linear spectrum has no term for the reset. The principled fix is a linearization that **includes** the reset as a jump map — saltation matrices, or Floquet theory of periodic-reset systems — whose spectrum would carry the reset frequency explicitly. This is the natural route to a rung between 5 and 6 that **could** reach the integer period.
- **Dynamic recalibration done right.** Followup A falsified a naive recalibration; a calibration that fits the time constant against impulse-response data, separately from the static input–output curve, would close the transfer-function **magnitude** bias even though it cannot close the period gap.
- **The full age-structured mesoscopic theory.** Quasi-renewal here uses the simplest single-integral form. The full age-structured equations [7], which carry phase information beyond a single hazard, would sharpen the staircase prediction — though the diagnostic principle predicts they still cannot close the ≈ 0.30 floor entirely.
- **The pre-filter as a delivered tool.** The sound $\rho(A)$ certificate (Section 3.3) is currently a result, not a tool. Composing it against a concrete Lustre encoding, so it actually shortens a Kind2 sweep, is a self-contained engineering deliverable.

7. Appendix A — Thread chronology

The compendium is organised by method, not by history. For the record, here is the research as it actually unfolded: seven threads over roughly seven weeks, each a phase-gated pipeline with its own advisor-facing note.

Thread 1 — the research notes and the analysis toolkit. The opening move: recognising that an LI&F neuron’s subthreshold dynamics are a first-order linear recurrence, hence open to the classical engineering toolkit — eigenvalues, Z -domain transfer functions, Bode plots, fixed-point analysis. Two notes were written, one for an engineering audience and one recasting the same ideas for computer scientists (eigendecomposition as PageRank, spectral radius as iterative-method convergence, fixed points as Markov-chain absorbing sets). Validated on a contrived 4-neuron winner-take-all network: the dominant eigenvector named the winner, the spectral radius predicted settling time, and every prediction was confirmed against PRISM and Monte Carlo. This is rungs 1–2, established. Its weakness — never tested on a real FCS archetype — is what thread 2 addressed.

Thread 2 — archetypes / spectral cartography. The honest test: do rung-1 spectral methods predict behaviour on the actual FCS contralateral and negative-loop archetypes? The answer was the two-regime split (Section 3.1.1): deterministic winner-take-all is combinatorial, reachability winner-take-all is spectral, and the spike-reset rule is what divides them. Pole placement for oscillation periods failed (dead end 3). This thread produced the diagnostic principle and the central two-regime figure.

Thread 3 — population / Wilson–Cowan. The lift to rung 3: replace each neuron with a population, average the reset into a smooth gain function, and recover the full bifurcation toolkit. The winner-take-all boundary became an exact pitchfork curve (99.96% classification), oscillation onset an exact Hopf curve, and the framework generalised to series, parallel, and positive-loop archetypes. Cross-validation against the discrete simulator exposed dead end 5 — the pitchfork is a lower bound, missing spike-timing lock-in. The note was rewritten (v2) for an FCS-native audience, with the discrete-to-continuous bridge table that Table 1 descends from.

Thread 4 — closed-form (Siegert, $H(\omega)$, quasi-renewal). Rungs 4–5: replace the hand-tuned gain curve with three physically-derived objects — the Siegert static rate, the Richardson transfer function, the Naud–Gerstner quasi-renewal mesoscopic. Siegert beat the hand-tuned curve; the transfer function’s self-consistency was exact at zero frequency; quasi-renewal recovered the finite-size structure. The cross-validation against discrete spike trains exposed the transfer-function **magnitude** bias — a calibration limitation that foreshadowed dead end 4.

Thread 5 — closed-form winner-take-all. The three rung-4/5 lenses brought onto the exact FCS Property 7 / Figure 10 staircase grid. Siegert is the 99.6%-recall envelope; the transfer-function latency gate is orthogonal to the staircase (dead end 6); quasi-renewal partially dissolves it, with a unimodal N -dependence quantified in a dedicated mechanism note. This thread produced the ≈ 0.70 Jaccard floor.

Thread 6 — multi-neuron winner-take-all. The same lenses extended from two neurons to $N > 2$. The two-neuron envelope did not survive: recall collapses with N , and the **inverse staircase** appears — discrete winner-take-all below the smooth-rate threshold (dead end 7). The continuous lens fails on both sides of the discrete boundary.

Thread 7 — closed-form negative loop. The three lenses on FCS Property 5 oscillation. Mean-field cannot produce the oscillation at all (it is a stable spiral); the transfer function gets the direction but the period is a structural $4 \times$ too long (dead end 4, with the falsified-recalibration

followup); quasi-renewal recovers the period exactly. Followups extended the result to a single-neuron renewal predictor and a three-neuron loop.

8. Appendix B — Per-thread phase scorecards

Each thread was a phase-gated pipeline; each phase tested one falsifiable hypothesis and recorded a verdict. The scorecards below put every verdict on one page. “Pass / partial / fail” are the threads’ own gate verdicts.

Thread 2 — archetypes / spectral cartography

Phase	Hypothesis tested	Verdict
0	Reproduce FCS semantics; build deterministic ground truth	pass
1a	Eigenvalue gap (scalar linearization) predicts WTA	fail
1b	Full windowed-integrator linearization fixes 1a	fail
1c	Spectral radius predicts WTA under reachability semantics	pass
2	Two-regime split transfers to the delayer-augmented topology	pass
3	Pole placement designs oscillation periods	fail

Thread 3 — population / Wilson–Cowan

Phase	Hypothesis tested	Verdict
0	Mean-field infrastructure: solver, Jacobian, fixed points	pass
1	Spectral gap predicts the WTA boundary (99.96%)	pass
2	Pitchfork and Hopf loci derivable in closed form	pass
3	Pole placement designs a target oscillation frequency	partial
4	The WC pitchfork is the discrete LI&F WTA boundary	fail
5	The framework generalises to series / parallel / positive loop	pass

Thread 4 — closed-form (Siegert / $H(\omega)$ / quasi-renewal)

Phase	Hypothesis tested	Verdict
0	Stochastic-LI&F bridge reaches the diffusion regime	pass
1	Siegert beats the hand-tuned gain curve	pass
2	Transfer-function self-consistency exact at zero frequency	pass
3	Transfer function predicts driven-response magnitude and phase	partial
4	Quasi-renewal recovers finite-size bistability structure	pass

Thread 5 — closed-form winner-take-all

Phase	Hypothesis tested	Verdict
0	Reproduce the FCS Property 7 staircase grid	pass
1	Siegert is a high-recall WTA envelope	pass
2	Transfer-function latency gate separates the staircase	partial
3	Quasi-renewal finite-size noise dissolves the staircase	partial

Thread 6 — multi-neuron winner-take-all

Phase	Hypothesis tested	Verdict
0	Reproduce FCS Property 7 at $N > 2$	pass
1	The Siegert envelope generalises beyond two neurons	partial
2	The transfer-function gate rescues the multi-neuron failure	partial
3	Quasi-renewal converges to mean-field at every N	pass

Thread 7 — closed-form negative loop

Phase	Hypothesis tested	Verdict
0	Reproduce FCS Property 5 oscillation	pass
1	Siegert fixed point predicts where oscillation occurs	partial
2	Transfer function predicts the oscillation period	partial
3	Quasi-renewal recovers the oscillation period	pass
A	Dynamic recalibration closes the period gap	fail
B	A single-neuron renewal predictor recovers the binary waveform	partial
C	The three-lens machinery generalises to a three-neuron loop	pass

Thread 1 (the foundational research notes and analysis toolkit) was not phase-gated; it is summarised in Section 7.

9. Appendix C — Glossary

Plain-language definitions of every technical term, for a reader without a differential-equations background. The section where each term is introduced in full is given in brackets.

Bifurcation a parameter value at which a system’s qualitative behaviour changes — a fixed point gains or loses stability, or splits. Bifurcations are the **boundaries** of behaviours in parameter space. (Section 2.4)

Bode plot a plot of the transfer-function magnitude $|H(\omega)|$ against driving frequency ω ; it shows how nimbly a system tracks fast input. (Section 2.3)

Diffusion approximation the assumption that a neuron’s input is noisy enough to be treated as a mean plus Gaussian fluctuations; the basis of the Siegert formula. (Section 2.5)

Eigenvalue for a linear map, a number λ by which some direction is rescaled each step; $|\lambda| < 1$ shrinks, $|\lambda| > 1$ grows. (Section 2.2)

Eigenvalue gap the margin by which the dominant eigenvalue stands above the rest; the spiking-network analogue of a Markov chain’s spectral gap. (Section 2.2)

Fixed point a state the dynamics map to itself — an equilibrium. A network can have several; which one is stable matters. (Section 2.1)

Hopf bifurcation the bifurcation at which a stable fixed point gives way to a sustained oscillation; the boundary of oscillatory behaviour. (Section 2.4)

Jaccard index an agreement score between two yes/no maps — the size of their overlap divided by the size of their union; 1 is perfect agreement. (Section 3.1)

Jacobian the matrix of the best linear approximation of a (nonlinear) dynamics at a fixed point; its eigenvalues decide local stability. (Section 2.2)

Linearization replacing a nonlinear rule by its best linear approximation near an operating point. (Section 2.1)

Mean-field a description that replaces a population of units by their average, exact in the limit of infinitely many units. (Section 2.4)

Pitchfork bifurcation the bifurcation at which one stable state splits into two — the winner-take-all boundary of mutual inhibition. (Section 2.4)

Pole placement choosing weights so that the linearized system has prescribed eigenvalues (“poles”) — inverse design. (dead end 3, Section 5)

Quasi-renewal a finite-population theory that tracks the age distribution (time since last spike) and adds $1/\sqrt{N}$ noise; rung 5. (Section 2.6)

Reachability whether a target behaviour is attainable from **some** admissible initial state — the question a model checker answers. (Section 3.3)

Siebert formula the closed-form expression for a noisy integrate-and-fire neuron’s firing rate, given its input mean and variance; rung 4. (Section 2.5)

Spectral radius ρ the largest eigenvalue magnitude; $\rho < 1$ is the condition for a linear system to settle. (Section 2.2)

Spike-reset rule the discontinuous jump of a neuron’s state after it fires; the one nonlinearity that every continuous method erases, and the subject of the diagnostic principle. (Section 1, Section 2.8)

Staircase the three diagonal blocks of synchronous-lock cells in the FCS winner-take-all grid — the integer-tick structure no rate theory reproduces. (Section 3.1)

Transfer function $H(\omega)$ the complex number giving a linear system’s gain and phase delay in response to input at frequency ω ; rung 2. (Section 2.3)

Wilson–Cowan equation the smooth differential equation for a neural population’s firing rate; the basis of rung 3. (Section 2.4)

Winner-take-all (WTA) the outcome where one neuron in a competing pair fires steadily and the other falls silent. (Section 3.1)

Compendium of the CogSpike DEQ research programme, threads 1–7 (deq/research_note, deq/archetypes, deq/population, deq/closed_form, deq/closed_form_wta, deq/closed_form_wta_multi, deq/closed_form_neg_loop). Figures are reproduced from the per-thread result directories; see deq/final/README.md for provenance. Companion paper draft: paper/continuous_lens.typ.

Bibliography

- [1] N. Zandian Jazi, E. De Maria, and C. Leturc, “A Tool for Formal Verification of Probabilistic Spiking Neural Networks via Weight-Discretized Quotient Abstractions,” in *Proc. International Conference on Artificial Neural Networks (ICANN)*, 2026.
- [2] E. De Maria *et al.*, “On the Use of Formal Methods to Model and Verify Neuronal Archetypes,” *Frontiers of Computer Science*, vol. 16, no. 3, p. 163404, 2020, doi: 10.1007/s11704-020-0029-6.
- [3] M. J. E. Richardson, “Firing-Rate Response of Linear and Nonlinear Integrate-and-Fire Neurons to Modulated Current-Based and Conductance-Based Synaptic Drive,” *Physical Review E*, vol. 76, no. 2, p. 21919, 2007, doi: 10.1103/PhysRevE.76.021919.
- [4] H. R. Wilson and J. D. Cowan, “Excitatory and Inhibitory Interactions in Localized Populations of Model Neurons,” *Biophysical Journal*, vol. 12, no. 1, pp. 1–24, 1972, doi: 10.1016/S0006-3495(72)86068-5.
- [5] A. J. F. Siegert, “On the First Passage Time Probability Problem,” *Physical Review*, vol. 81, no. 4, pp. 617–623, 1951, doi: 10.1103/PhysRev.81.617.

- [6] R. Naud and W. Gerstner, “Coding and Decoding with Adapting Neurons: A Population Approach to the Peri-Stimulus Time Histogram,” *PLoS Computational Biology*, vol. 8, no. 10, p. e1002711, 2012, doi: 10.1371/journal.pcbi.1002711.
- [7] T. Schwalger, M. Deger, and W. Gerstner, “Towards a Theory of Cortical Columns: From Spiking Neurons to Interacting Neural Populations of Finite Size,” *PLoS Computational Biology*, vol. 13, no. 4, p. e1005507, 2017, doi: 10.1371/journal.pcbi.1005507.